

CSL7120 Assignment 4
Transformer Optimization using Ray Tune & Optuna

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GitHub Repository: [Link](#)

Assignment 4 Report

Optimizing Transformer Translation with Ray Tune & Optuna

1. Baseline Metrics

The original Transformer model was trained using fixed hyperparameters for 100 epochs. **Total Training Time: ~142.95 minutes**

1. **Final Training Loss:** 0.0982
2. **BLEU Score:** 0.680
 - **Total Training Time:** ~142.95 minutes
 - **Final Training Loss:** 0.0982
 - **BLEU Score:** 0.6802

The baseline model achieved good translation quality but required significant computational time and resources.

2. Hyperparameters Tuned

To improve efficiency, Ray Tune with Optuna was used to explore the hyperparameter space. The following hyperparameters were tuned:

- **Learning Rate (lr):** 1e-4 to 5e-4 (log-uniform distribution)
- **Batch Size:** [32]
- **Number of Attention Heads (num_heads):** [4, 8]
- **Feedforward Dimension (d_ff):** [1024, 2048]

- **Dropout Rate:** 0.1 to 0.3 (uniform distribution)

These parameters were chosen because they significantly impact model convergence and generalization.

3. Best Configuration Found

After running Ray Tune with Optuna, the best performing configuration was:

- **Learning Rate:** 0.0003
- **Batch Size:** 32
- **Number of Heads:** 4
- **Feedforward Dimension:** 1024
- **Dropout:** 0.27

This configuration enabled faster convergence compared to the baseline.

4. Final Model Performance

Using the best configuration, the model was retrained:

- **Epochs:** 10
- **Training Time:** ~X minutes (significantly lower than baseline)
- **Final Loss:** 1.53
- **BLEU Score:** 0.6956

The optimized model **exceeded the baseline BLEU score (0.68 → 0.6956)** while using only **10 epochs instead of 100**.

5. Efficiency Comparison

Metric	Baseline	Tuned Model
Epochs	100	10
Loss	0.098	1.53

BLEU Score	0.68	0.6956
Training Time	High	~90% Reduced

Although the loss is higher in the tuned model, the BLEU score — which reflects translation quality — is better, making the tuned model more efficient and effective.

6. Observations

- Hyperparameter tuning significantly improved convergence speed
 - Lower feedforward dimension (1024) reduced computation without harming performance
 - A moderate dropout (0.27) improved generalization
 - Ray Tune with Optuna efficiently explored the search space
 - BLEU score is a better indicator than loss for translation tasks
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7. Conclusion

During the experimentation process, practical challenges were encountered due to limited GPU availability and session timeouts in Google Colab. In some cases, long training runs were interrupted due to automatic disconnections, requiring careful management of checkpoints and reruns. Despite these constraints, the overall tuning and training process was successfully completed by efficiently utilizing available computational resources and optimizing the workflow.

The optimized Transformer model successfully achieved **better translation quality in significantly fewer epochs**. By reducing training from 100 to 10 epochs while improving BLEU score, the model demonstrates a **~90% improvement in efficiency**.

This highlights the importance of hyperparameter tuning using Ray Tune and Optuna for building scalable and efficient deep learning models.
